

PROBLEM STATEMENT #21 | Smart Cities, Transport & Logistics**EV Charging Station Reservation System****1. Problem Context & Overview**

Electric vehicle drivers require a navigation and reservation portal to locate compatible charging sockets (CCS, CHAdeMO, Type 2), reserve guaranteed time slots, and calculate dynamic power rates.

Target Stakeholders / Actors: *EV Driver, Station Operator*

2. Sample Functional Requirement (FR) Guideline**Sample Requirement: FR-001 [Priority: High]**

Description: The system shall match the driver vehicle connector type with available station chargers and reserve a 30-minute charging window.

Sample Acceptance Criteria: Pass: Incompatible connector plugs hidden from driver; Fail: Two vehicles booked for same socket concurrently.

3. Sample Non-Functional Requirement (NFR) Guideline**Sample Requirement: NFR-001 [Type: Performance & Security]**

Description: The system shall automatically release locked sockets and apply a penalty if the driver fails to plug in within 10 minutes of slot start.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]**• Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

• UML Use-Case Diagram:

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

• Use-Case Flow Specification:

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.